Assignment 3

Q1(a) - Points 5 (1,0) & 6 (0,1) are the -1 points close to +1 Group. Both lie on the line $x_1 + x_2 = 1$

- point 1 (1,1) is the +1 point close to the -1 group, on the parallel line $x_1 + x_2 = 2$

- points 2 (1,2) & 3 (2,1) are on $x_1 + x_2 = 3$ and point 4 (0,0) on $x_1 + x_2 = 0$ sit further back from the gap. So, they are not support vectors

Ans: Support vectors are points 1 (1,1), 6 (0,1) and 5 (1,0)

Updated the template: (assignment3_template_updwith_SpportVctrs.py) ↓ Backup.
✓ assignment3_template.py

Under the function:

identify_support_vectors()

return [1, 5, 6]

Q1(b) Boundary & Margin:

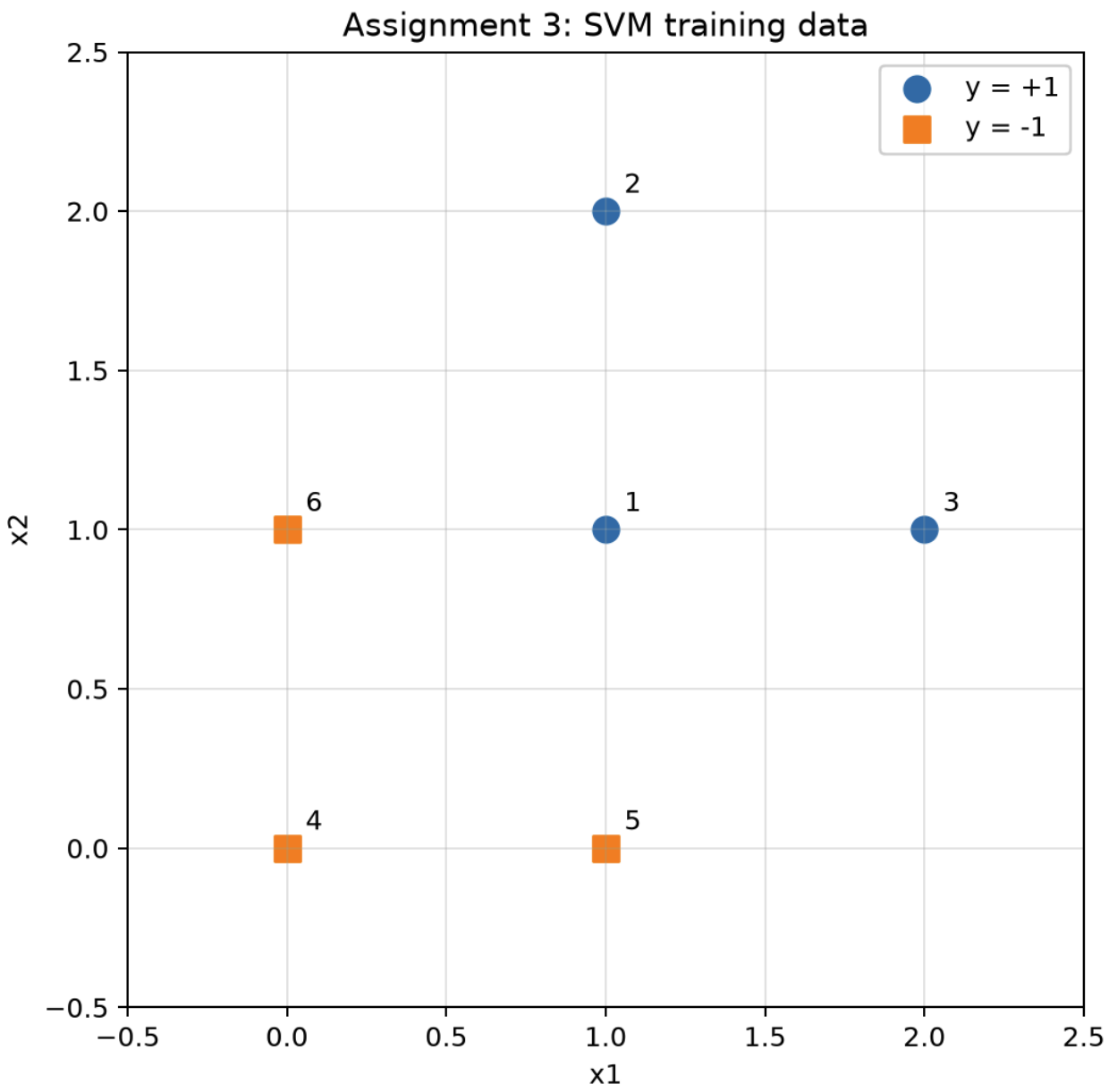
For hard margin SVM,

Support vectors satisfy:

$w \cdot x + b = +1$ (positive ones)

$w \cdot x + b = -1$ (-ve ones)

where $w = (w_1, w_2)$


Assignment 3: SVM training data
y = +1
y = -1
1
2
3
4
5
6
x1
x2
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−0.5
0.0
0.5
1.0
1.5
2.0
2.5

Support Vectors:

1 (1,1) $y = +1 \Rightarrow w_1(1) + w_2(1) + b = +1 \Rightarrow w_1 + w_2 + b = 1$ — ①

5 (1,0) $y = -1 \Rightarrow w_1(1) + w_2(0) + b = -1 \Rightarrow w_1 + b = -1$ — ②

6 (0,1) $y = -1 \Rightarrow w_1(0) + w_2(1) + b = -1 \Rightarrow w_2 + b = -1$ — ③

From Eqns ② & ③

$w_1 = w_2$

Say $w_1 = w_2 = w$

$\Rightarrow$ pt 5 ② $\Rightarrow w + b = -1$

pt 6 ③ $\Rightarrow w + b = -1$

pt 1 ① $\Rightarrow w + w + b = 1 \Rightarrow 2w + b = 1$ — ④

Solving the Eqns for 'w'

From ③ & ③ $w + b = -1 \Rightarrow 2w + 2b = -2$

From ④ $2w + b = 1 \Rightarrow -2w - b = -1$

$b = -3$

$w + b = -1$

$\Rightarrow w = -1 - b = -1 - (-3) = -1 + 3 = 2$

$\therefore \boxed{w = 2}$ & $\boxed{b = -3}$

$\therefore$ The decision Boundary:

$2x_1 + 2x_2 - 3 = 0$

$\Rightarrow x_1 + x_2 = 3/2 \Rightarrow \boxed{x_1 + x_2 = 1.5}$

Geometric Mean

$$\text{margin} = \frac{1}{\|w\|}$$

$$\|w\| = \sqrt{w_1^2 + w_2^2}$$

$w_1 = 2$ & $w_2 = 2$

$$\Rightarrow \|w\| = \sqrt{2^2+2^2} = \sqrt{8} = 2\sqrt{2} = 2 \times (1.414) = 2.828$$

Ans:

Geometric margin $= \frac{1}{\|w\|} = \frac{1}{2.828} = 0.3536$

Ans:

Full street width $= \frac{2}{\|w\|} = \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}} = 0.7071$

Updated the template: assignment3_template.py

under: desire_srm_separator ( )

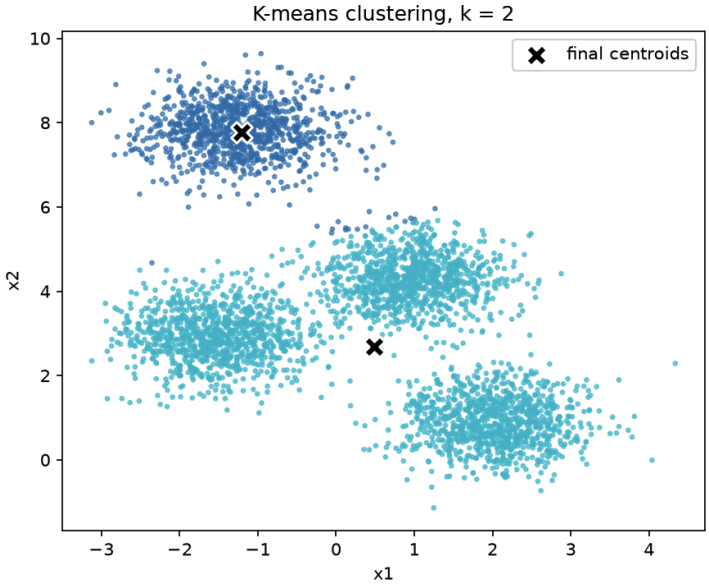
Q2) K-means

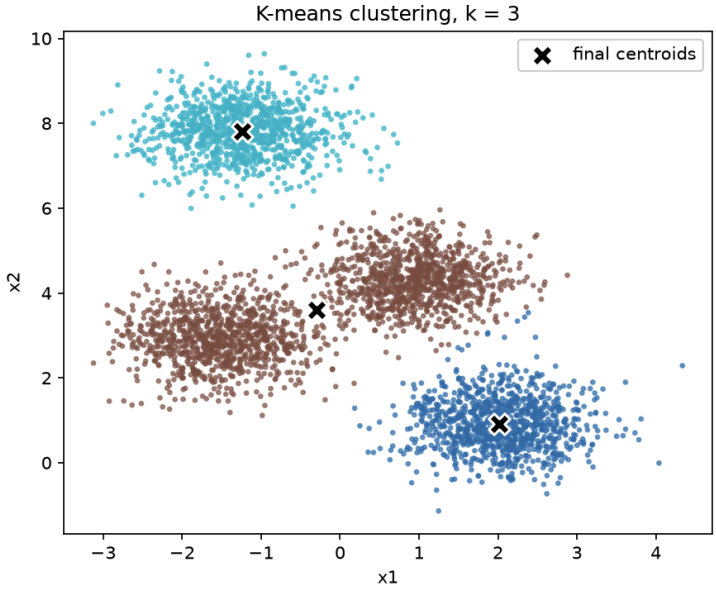
- Updated the kmeans_template.py
- (% python kmeans_template.py --seed 0 --max-iterations 100)
- generated the plots saved under results/kmeans

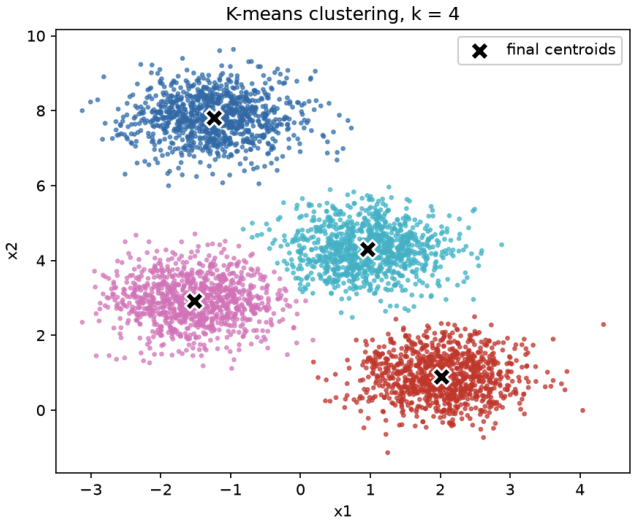
kmeans_k*.png

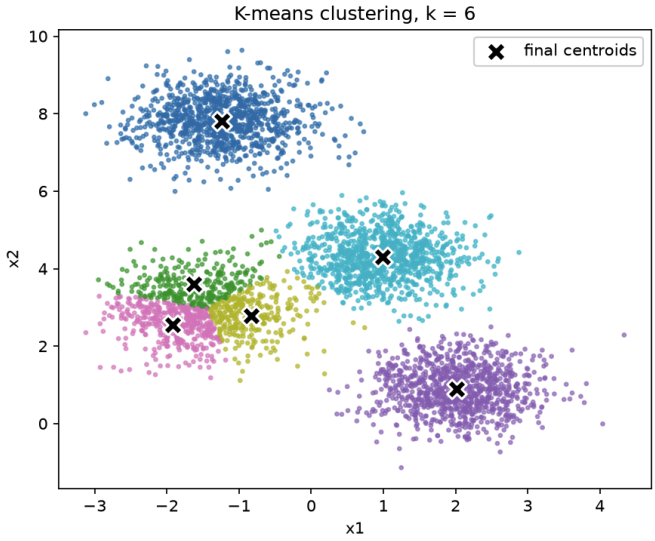
Looking at the 5 plots and the numbers:

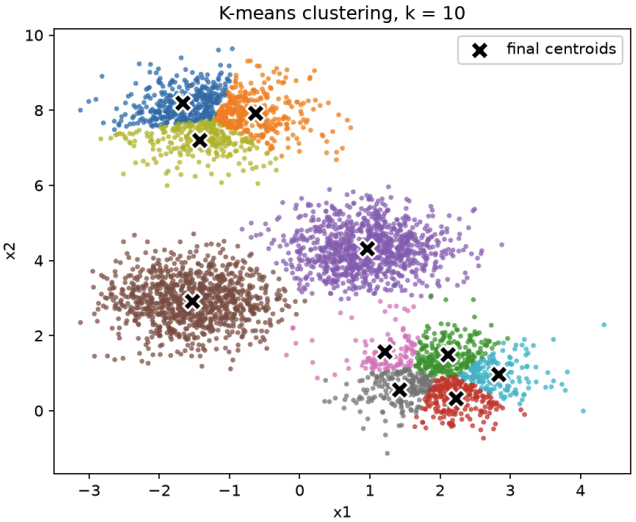
1. K = 2, 3 ⇒ too few clusters. K-means merges true groups.
2. K = 4 ⇒ matches 4 generating groups. Centroid is at the blob centers. Sizes are almost 1000 each.
3. K = 6, 10 ⇒ too many clusters. K-means splits real groups into arbitary pieces.


K-means clustering, k = 2
final centroids
x1
x2


K-means clustering, k = 3
final centroids
x1
x2


K-means clustering, k = 4
final centroids
x1
x2


K-means clustering, k = 6
final centroids
x1
x2


K-means clustering, k = 10
final centroids
x1
x2

4. Sum of Squared Errors (SSE) drops as k increases.

Q3) Noisy labels & K-NN Repair

Updating the template : noisy_classification_template.py

and running for 20 repeats:

(% python noisy_classification_template.py --repeats 20 --seed 0)

Mean baseline accuracy : 0.828

Mean K-NN repair accuracy : 0.8644

Difference (K-NN − Baseline) : +0.0364 = ~3.6% points

- It helped since the label noise is random & scattered with the true classes spatially clustered. The majority vote fixes it. SVM then sees a smoother consistent labeling & cleaner boundary
- Sometimes it can hurt near the overlap between the two Gaussians. When neighbors are mixed the vote can overwrite correct labels. Noise is also asymmetric.

noise_comparison

trial	baseline_accuracy	knn_accuracy
1	0.8345	0.8615
2	0.833	0.872
3	0.844	0.8675
4	0.8445	0.87
5	0.8395	0.8725
6	0.841	0.8715
7	0.8405	0.869
8	0.793	0.848
9	0.824	0.8615
10	0.8425	0.8715
11	0.822	0.86
12	0.819	0.8645
13	0.847	0.8645
14	0.8415	0.8715
15	0.816	0.8715
16	0.8285	0.8485
17	0.8235	0.8695
18	0.843	0.87
19	0.784	0.8335
20	0.798	0.869